

第3章 集合

集合

一、概念

1. 集合中的元素可以是有限的,也可以是无限的。基数 $\text{card} S = |S| = \underline{\quad}$ (个)
 $|S|$ 表示集合 S 中元素的个数

2. 常用集合符号

$I_m (m \geq 1)$: 有限个正整数的集合 $\{1, 2, \dots, m\}$

$N_m (m \geq 0)$: 有限个自然数的集合 $\{0, 1, 2, \dots, m\}$

N : 自然数集合

I^+ : 正整数集合

I : 整数集合

P : 素数集合

Q : 有理数集合 $\{\frac{i}{j}, i, j \text{ 均为整数且 } j \neq 0\}$

R : 实数集合

C : 复数集合 $\{a+bi, a, b \text{ 可为实数, } i = \sqrt{-1}\}$

二、表示方法

① 枚举法

如 $A = \{T, F\}$, $B = \{1, 2, 3, 4\}$, $C = \{a+1, a+2\}$, $D = \{a, A, b, B, c, C\}$

② 谓词公式描述法

如 $S = \{x \mid P(x)\}$

大于10的整数集合 $S = \{x \mid x \in I \wedge x > 10\}$

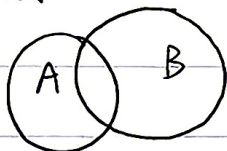
偶数集合 $S = \{x \mid y (y \in I \wedge x = 2y)\}$

$S = \{1, 2, 3, 4\} = \{x \mid x \in I \wedge (1 \leq x \leq 4)\}$

$S = \{F, T\} = \{x \mid x = T \vee x = F\}$

$S = \{1, 4\} = \{x \mid (x^2 - 5x + 4 = 0)\}$

③ 文氏图法



三个特殊的集合:

① 全集 $E = \{x | p(x) \vee \neg p(x)\}$

② 空集 $\phi = \{x | p(x) \wedge \neg p(x)\} = \{ \}$

注: ϕ 表示空集, 是无元素的集合; $\{\phi\}$ 是以 ϕ 作为元素的集合

③ 集合族 如 $A = \{\{a\}, \{b, c\}, \{d\}\}$

三. 关系

① 当且仅当 A 和 B 具有同样元素时, $A=B$.

规定 $(A=B) \Leftrightarrow (\forall x)(x \in A \Leftrightarrow x \in B)$

② A 包含于 B $A \subseteq B$ 或 B 包含 A $B \supseteq A$

$A \subseteq B \Leftrightarrow B \supseteq A \Leftrightarrow (\forall x)(x \in A \rightarrow x \in B)$

$A=B$ 当且仅当 $A \subseteq B$ 和 $B \subseteq A$

证明:

$(A=B) \Leftrightarrow (\forall x)((x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A))$

$\Leftrightarrow (\forall x)(x \in A \rightarrow x \in B) \wedge (\forall x)(x \in B \rightarrow x \in A)$

$\Leftrightarrow (A \subseteq B) \wedge (B \subseteq A)$

A 真包含于 B $A \subset B \Leftrightarrow (A \subseteq B \wedge A \neq B)$

$A \subseteq B \wedge A \neq B \Leftrightarrow (\forall x)(x \in A \rightarrow x \in B) \wedge (\exists x)(x \in B \wedge x \notin A)$

$\subseteq$ 指集合间的关系, $\in$ 指元素与集合间的关系

$S = \{a, \{b\}\}$, 则 $a \in S$, $\{b\} \in S$

③ 设有空集 ϕ 和任一集合 A, 则 $\phi \subseteq A$

证明: $\because \phi$ 中无元素 $\therefore x \in \phi$ 为假 $\therefore (\forall x)(x \in \phi \rightarrow x \in A)$ 为真

④ 当且仅当 $A \subseteq B$ 和 $B \subseteq A$ 时, 才有 $A=B$

证明:

(必要性) $(A \subseteq B) \wedge (B \subseteq A)$

(充分性) $A=B$

$\Leftrightarrow (\forall x)(x \in A \rightarrow x \in B) \wedge (\forall x)(x \in B \rightarrow x \in A) \Leftrightarrow (\forall x)(x \in A \Leftrightarrow x \in B)$

①

$\Leftrightarrow (\forall x)((x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)) \Leftrightarrow (\forall x)((x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A))$

$\Leftrightarrow (\forall x)(x \in A \Leftrightarrow x \in B) \Leftrightarrow (\forall x)(x \in A \rightarrow x \in B) \wedge (\forall x)(x \in B \rightarrow x \in A)$

$\Leftrightarrow A=B \Leftrightarrow (A \subseteq B) \wedge (B \subseteq A)$

⑤ 若 $A \subseteq B, B \subseteq C$, 则 $A \subseteq C$ (" $\subseteq$ " 改为 " $\subset$ " 亦成立)

⑥ ϕ 是唯一的

证明: 设有 2 个空集 ϕ_1 和 ϕ_2 $\forall \phi$ 是任何集合的子集

$\therefore (\phi_1 \subseteq \phi_2 \wedge \phi_2 \subseteq \phi_1) \Leftrightarrow \phi_1 = \phi_2$ 故空集是唯一的

四、幂集和索引集合

由 A 所有子集为元素组成的集合, 称为 A 的幂集, 记作 $P(A)$.

若 $A = \phi$, 则 $P(A) = \{\phi\}$

若 $A = \{a\}$, 则 $P(A) = \{\phi, \{a\}\}$

若 $A = \{1, 2\}$, 则 $P(A) = \{\phi, \{1\}, \{2\}, \{1, 2\}\}$

★ 若 $A = \phi$, 求 $P(P(A))$ 的幂集

解: $\because A = \phi \therefore P(A) = \{\phi\}$

$P(P(A)) = \{\phi, \{\phi\}\}$

$P(\{\phi, \{\phi\}\}) = \{\phi, \{\phi, \{\phi\}\}, \{\phi\}, \{\{\phi\}\}\}$

$P(A) = 2^A = \{x | x \subseteq A\}$

证明: $C_n^k = \frac{n(n-1)\dots(n-k+1)}{k!}$

$\forall \phi \subseteq A$

$\therefore N = 1 + C_n^1 + C_n^2 + \dots + C_n^k + \dots + C_n^n$
 $= \sum_{k=0}^n C_n^k$

$\forall (x+y)^n = \sum_{k=0}^n C_n^k \cdot x^k \cdot y^{n-k}$ (令 $x=1, y=1$)

则 $\sum_{k=0}^n C_n^k = 2^n$

索引集合

如 $\phi = \{\} = B_{00}, \{a\} = B_{10}, \{b\} = B_{01}, \{a, b\} = B_{11}$

$P(S) = \{B_{00}, B_{01}, B_{10}, B_{11}\} = \{B_{ij} | i, j \in J\}$ ($J = \{00, 01, 10, 11\}$, 称为索引(指标集))

已知 $S = \{a_1, a_2, a_3, a_4, a_5, a_6\}$, 有 2^6 个子集, 则可分别表示为 $B_0, B_1, \dots, B_{2^6-1}$

$B_{000111} = \{a_4, a_5, a_6\} = B_7$

$B_{001111} = \{a_3, a_4, a_5, a_6\} = B_{15}$

$B_{010110} = \{a_2, a_4, a_5\} = B_{22}$

⑤ 证明: $A \subseteq B \wedge B \subseteq C$

$\Leftrightarrow (\forall x)(x \in A \Rightarrow x \in B) \wedge (A \neq B) \wedge (\forall x)((x \in B \Rightarrow x \in C) \wedge (B \neq C))$

$\Leftrightarrow (\forall x)(x \in A \Rightarrow x \in B) \wedge (\forall x)(x \in B \Rightarrow x \in C) \wedge (A \neq B) \wedge (B \neq C)$

$\Leftrightarrow (\forall x)((x \in A \Rightarrow x \in B) \wedge (x \in B \Rightarrow x \in C)) \wedge (A \neq C)$

$\Leftrightarrow (\forall x)(x \in A \Rightarrow x \in C) \wedge (A \neq C)$

$\Leftrightarrow A \subseteq C$

五、运算

① 交运算 (可交换、可结合)

$$A \cap B = \{x \mid x \in A \wedge x \in B\} \quad \text{若 } A \cap B = \emptyset, \text{ 则称 } A, B \text{ 不相交}$$

$$A \cap A = A, \quad A \cap \emptyset = \emptyset, \quad A \cap E = A, \quad A \cap B = B \cap A, \quad (A \cap B) \cap C = A \cap (B \cap C)$$

$$\text{若 } A \subseteq B, \text{ 则 } A \cap C \subseteq B \cap C$$

$$\text{若 } A \cap B \subseteq A \text{ 则 } A \cap B \subseteq B$$

证明: $\forall x \in A \cap C$, 有 $x \in A$ 且 $x \in C$

又: $A \subseteq B$, $\therefore x \in B$ 且 $x \in C$, 故 $x \in B \cap C$ 因此 $A \cap C \subseteq B \cap C$

② 并运算 (可交换、可结合)

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

$$A \cup A = A, \quad A \cup \emptyset = A, \quad A \cup E = E, \quad A \cup B = B \cup A, \quad (A \cup B) \cup C = A \cup (B \cup C)$$

$$\text{若 } A \subseteq B, C \subseteq D, \text{ 则 } A \cup C \subseteq B \cup D$$

$$\text{若 } A \subseteq A \cup B, \text{ 则 } B \subseteq A \cup B$$

证明: $\forall x \in A \cup C$, 有 $x \in A$ 或 $x \in C$

若 $x \in A$, 由 $A \subseteq B$, 得 $x \in B$, 故 $x \in B \cup D$

若 $x \in C$, 由 $C \subseteq D$, 得 $x \in D$, 故 $x \in B \cup D$

$\Rightarrow$ 故 $A \cup C \subseteq B \cup D$

分配律: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

以第1条为例, 证明:

$$\text{设 } S = A \cap (B \cup C), \quad T = (A \cap B) \cup (A \cap C)$$

若 $x \in S$, 则 $x \in A$ 且 $x \in B \cup C$, 即 $x \in A$ 且 $x \in B$ 或 $x \in A$ 且 $x \in C$

即 $x \in A \cap B$ 或 $x \in A \cap C$, 即 $x \in T$ $\therefore S \subseteq T$

若 $x \in T$, 则 $x \in A \cap B$ 或 $x \in A \cap C$, 即 $x \in A$ 且 $x \in B$ 或 $x \in A$ 且 $x \in C$

即 $x \in A$ 且 $x \in B \cup C$, 即 $x \in S$ $\therefore T \subseteq S$

故 $S = T$

若 $A \subseteq B$, 则 $A \cap B = A$

若 $A \subseteq B$, 则 $A \cup B = B$

$$A \cup (A \cap B) = A$$

证明: 左边 = $(A \cap E) \cup (A \cap B)$
 $= A \cap (E \cup B) = A$

$$A \cap (A \cup B) = A$$

证明: 左边 = $(A \cup A) \cap (A \cup B)$
 $= A \cap (A \cup B) = A$

$A \subseteq B$ 当且仅当 $A \cup B = B$ 或 $A \cap B = A$

证明: 以 $A \cup B = B$ 为例

需证明 $\begin{cases} A \subseteq B \rightarrow A \cup B = B \\ A \cup B = B \rightarrow A \subseteq B \end{cases}$

② [必要性] 对 $\forall x \in A \cup B$, 有 $x \in A$ 或 $x \in B$

$x \because A \subseteq B \quad \therefore x \in B \quad \text{可得 } A \cup B \subseteq B$

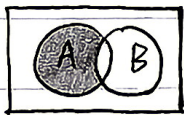
$x \because B \subseteq A \cup B$

故 $A \cup B = B$

[充分性] $\because A \cup B = B$, $x \because A \subseteq A \cup B$, 即 $A \subseteq B$.

$$\begin{matrix} A \subseteq B & \Leftrightarrow & A \cup B = B & \Leftrightarrow & A \cap B = A \\ \textcircled{1} & & \textcircled{2} & & \textcircled{3} \end{matrix}$$

③ 相对补、绝对补



B对A的相对补

$$\begin{aligned} A - B &= \{x \mid x \in A \wedge x \notin B\} \\ &= \{x \mid x \in A \wedge \neg x \in B\} \end{aligned}$$

$$A - B \subseteq A, A - \phi = A, A \cap (B - A) = \phi, A \cup (B - A) = A \cup B$$



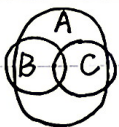
A的绝对补(称补)

$$\sim A = E - A = \{x \mid x \in E \wedge x \notin A\}$$

$$= \{x \mid x \notin A\} = \{x \mid \neg(x \in A)\}$$

$$\sim(\sim A) = A, \sim E = \phi, \sim \phi = E, A \cup \sim A = E, A \cap \sim A = \phi$$

$$A - (B \cup C) = (A - B) \cap (A - C)$$



$$A - (B \cap C) = (A - B) \cup (A - C)$$

若 $A \subseteq B$ 和 $C \subseteq D$, 则 $A \cap C \subseteq B \cap D$

证明: 对任意 $x \in A \cap C$, 则 $x \in A \wedge x \in C$

$x \because A \subseteq B$ 和 $C \subseteq D$ 故 $x \in B \wedge x \in D$

即 $x \in B \cap D$

则 $A \cap C \subseteq B \cap D$

$$\sim(A \cup B) = \sim A \cap \sim B$$

$$\begin{aligned} \text{证明: } \sim(A \cup B) &= \{x | x \in \sim(A \cup B)\} \\ &= \{x | x \notin A \cup B\} \\ &= \{x | (x \notin A) \wedge (x \notin B)\} \\ &= \{x | (x \in \sim A) \wedge (x \in \sim B)\} \\ &= \sim A \cap \sim B \end{aligned}$$

$$\sim(A \cap B) = \sim A \cup \sim B$$

$$\begin{aligned} \text{证明: } \sim(A \cap B) &= \{x | x \in \sim(A \cap B)\} \\ &= \{x | x \notin A \cap B\} \\ &= \{x | (x \notin A) \vee (x \notin B)\} \\ &= \{x | (x \in \sim A) \vee (x \in \sim B)\} \\ &= \sim A \cup \sim B \end{aligned}$$

$$A - B = A \cap \sim B$$

$$\begin{aligned} \text{证明: 对 } \forall x \in (A - B) \\ \text{则 } x \in A \text{ 且 } x \notin B, \text{ 即 } x \in A \text{ 且 } x \in \sim B \\ \text{即 } x \in A \cap x \in \sim B, x \in (A \cap \sim B) \text{ 故 } A - B \subseteq A \cap \sim B \\ \text{对 } \forall x \in (A \cap \sim B) \\ \text{则 } x \in A \text{ 且 } x \in \sim B, \text{ 即 } x \in A \text{ 且 } x \notin B, \\ \text{故 } x \in (A - B), \text{ 故 } A \cap \sim B \subseteq A - B \\ \text{综上, 得 } A - B = A \cap \sim B \end{aligned}$$

$$A - B = A - (A \cap B)$$

$$\begin{aligned} \text{证明: 对 } \forall x \in (A - (A \cap B)) \\ \text{则 } x \in A \text{ 且 } x \notin A \cap B, \text{ 即 } x \in A \text{ 且 } x \notin (A \cap B) \\ \text{即 } x \in A \cap x \notin (A \cap B), x \in (A \cap (\sim A \cup \sim B)) \\ x \in ((A \cap \sim A) \cup (A \cap \sim B)), x \in (A \cap \sim B) \\ \text{即 } x \in A \text{ 且 } x \notin B, \text{ 故 } x \in (A - B) \\ \text{可得 } A - (A \cap B) \subseteq A - B \\ \text{同理可推 } A - B \subseteq A - (A \cap B) \end{aligned}$$

$$A \cap (B - C) = (A \cap B) - (A \cap C)$$

$$\begin{aligned} \text{证明: } \because A \cap (B - C) \\ &= A \cap (B \cap \sim C) = A \cap B \cap \sim C \\ \text{又: } (A \cap B) - (A \cap C) \\ &= (A \cap B) \cap \sim (A \cap C) \\ &= (A \cap B) \cap (\sim A \cup \sim C) \\ &= (A \cap B \cap \sim A) \cup (A \cap B \cap \sim C) \\ &= A \cap B \cap \sim C \\ \text{故两边相等} \end{aligned}$$

$$B = \sim A \text{ 当且仅当 } A \cup B = E \text{ 且 } A \cap B = \emptyset$$

证明: [必要性]

$$\because B = \sim A \therefore A \cup B = A \cup \sim A = E$$

$$A \cap B = A \cap \sim A = \emptyset$$

③ [充分性]

$$\because A \cup B = E \text{ 且 } A \cap B = \emptyset$$

$$B = B \cap E = B \cap (A \cup \sim A)$$

$$= (A \cap B) \cup (\sim A \cap B) = \emptyset \cup (\sim A \cap B)$$

$$= (\sim A \cap A) \cup (\sim A \cap B)$$

$$= \sim A \cap (A \cup B) = \sim A \cap E$$

$$= \sim A$$

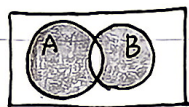
感悟: 证明“当且仅当”时3种类型

①②③

④ 对称差 (环和) (可交换、可结合)

$$(A \oplus B) = (A \cup B) - (A \cap B)$$

$$A \oplus B = (A - B) \cup (B - A) = (A \cap \sim B) \cup (B \cap \sim A) = (A \cup B) \cap (\sim A \cup \sim B) = (A \cup B) \cap \sim (A \cap B)$$



$$A \oplus B = B \oplus A, (A \oplus B) \oplus C = A \oplus (B \oplus C), A \oplus A = \emptyset, A \oplus \emptyset = A$$

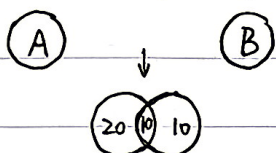
$$A \cap (B \oplus C) = (A \cap B) \oplus (A \cap C)$$

$$\sim A \oplus \sim B = (\sim A - \sim B) \cup (\sim B - \sim A) = (\sim A \cap B) \cup (\sim B \cap A) = (B - A) \cup (A - B) = A \oplus B$$

$$x \in (A \oplus B) \Leftrightarrow x \in \{x \mid x \in A \vee x \in B\}$$

六. 计数

1. 有100个糖, 30根有黄色, 20根有紫色, 10根上既有黄色又有紫色



$$|A - B| = |A| - |A \cap B| = 30 - 10 = 20$$

$$|B - A| = |B| - |B \cap A| = 20 - 10 = 10$$

有多少是黄的或紫的? $|A \cup B| = |A| + |B| - |A \cap B| = 40$

2. 包含排斥原理

$$|A \cup B| = |A| + |B| - |A \cap B|$$

证明: 当 $A \cap B = \emptyset$ 时, 成立

$$\text{当 } A \cap B \neq \emptyset \text{ 时, } |A| = |A \cap \sim B| + |A \cap B|$$

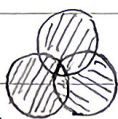
$$|B| = |B \cap \sim A| + |A \cap B|$$

$$\therefore |A| + |B| = |A \cap \sim B| + |B \cap \sim A| + 2|A \cap B|$$

$$\text{又: } |A \cup B| = |A \cap B| + |A \cap \sim B| + |B \cap \sim A|$$

$$\text{故 } |A \cup B| = |A| + |B| - |A \cap B|$$

推广: $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$



$$|S_1 \cup S_2 \cup \dots \cup S_n| = \sum_{i=1}^n |S_i| - \sum_{1 \leq i < j \leq n} |S_i \cap S_j| + \sum_{1 \leq i < j < k \leq n} |S_i \cap S_j \cap S_k| - \dots + (-1)^{n+1} |S_1 \cap S_2 \cap \dots \cap S_n|$$

$$|S_1 \cap S_2 \cap \dots \cap S_n| = |S_1| - \sum_{i=1}^n |S_i| + \sum_{1 \leq i < j \leq n} |S_i \cap S_j| - \sum_{1 \leq i < j < k \leq n} |S_i \cap S_j \cap S_k| + \dots + (-1)^n |S_1 \cap S_2 \cap \dots \cap S_n|$$

24个人, 每人至少会1门外语

英: 13人 俄: 5人 德: 10人 法: 9人

会英俄: 2人 会英德: 4人 会英法: 4人 会法德: 4人

会俄不会法和德.

求同时会英、法、德3种, 和只会英、法、德中1种的人数.

解:

设 A, B, C, D 分别表示会英、俄、德、法语的人的集合

设3种都会的人为 x , 只会三者之一的分别为 y_1, y_2, y_3

由题意, 只会俄语的有 $5-2=3$ (人)

$$\therefore |A \cup C \cup D| = 21; |A| = 13, |C| = 10, |D| = 9$$

$$|A \cap C| = 4, |A \cap D| = 4, |C \cap D| = 4$$

$$|A \cup C \cup D| = |A| + |C| + |D| - |A \cap C| - |A \cap D| - |C \cap D| + |A \cap C \cap D|$$

$$21 = 13 + 10 + 9 - 4 \times 3 + x \quad \text{解得 } x = 1$$

$$\therefore |A - (C \cup D)| = |A| - (|A \cap C| + |A \cap D|) + |A \cap C \cap D| = 13 - (4 + 4) + 1 = 6$$

6人中2人会日语, $\therefore$ 只会英: $6-2=4$ (人)

$$|C - (A \cup D)| = 9 - (4 + 4) + 1 = 2 \text{ (人)}$$

$$|D - (A \cup C)| = 10 - (4 + 4) + 1 = 3 \text{ (人)}$$



例. 学校1807个新生中有453人选了计算机, 567人选了数学, 299人同时选了计算机和数学.

(1) 多少学生选了计算机或数学?

用 S 表示全体新生, 用 A, B 分别表示选了计算机和数学的学生集合

$$|A \cup B| = |A| + |B| - |A \cap B| = 453 + 567 - 299 = 721 \text{ (人)}$$

(2) 多少学生既没选计算机又没选数学?

$$\begin{aligned} |A^c \cap B^c| &= |S - (A \cup B)| = |S| - |A \cup B| = |S| - (|A| + |B| - |A \cap B|) \\ &= 1807 - 721 = 1086 \text{ (人)} \end{aligned}$$

例. 试计算在 $\{1, 2, 3, \dots, 1000\}$ 中有多少元素至少能被5, 6, 8这三个数中的一个整除.

解: 设 A, B, C 分别表示1000以内能被5, 6, 8整除的正整数集合

$$|A| = \left\lfloor \frac{1000}{5} \right\rfloor = 200, \quad |B| = \left\lfloor \frac{1000}{6} \right\rfloor = 166, \quad |C| = \left\lfloor \frac{1000}{8} \right\rfloor = 125$$

又: 一个数能被若干个数同时整除当且仅当其可被最小公倍数整除

$$|A \cap B| = \left\lfloor \frac{1000}{30} \right\rfloor = 33, \quad |A \cap C| = \left\lfloor \frac{1000}{40} \right\rfloor = 25, \quad |B \cap C| = \left\lfloor \frac{1000}{24} \right\rfloor = 41$$

$$|A \cap B \cap C| = \left\lfloor \frac{1000}{120} \right\rfloor = 8$$

$$\therefore |A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C| = 400 \text{ (个)}$$